



Mitigating Resonant OOR Development in Rail Transit: Multi Body Dynamics Investigation of New Fleet

Anbo Wang, Senior Consultant
Hatch

B O S T O N



What is OOR (Out-of-Round)?

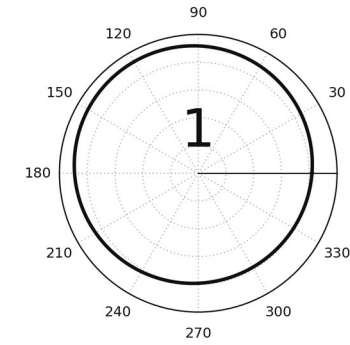
- Wheel Out-of-Roundness (OOR) is a condition in which the radius of a railway wheel varies around its circumference, causing deviations from the ideal circular shape and generating periodic or impact wheel-rail forces during wheel rotation. OOR may occur as localized defects such as wheel flats or as periodic irregularities such as wheel polygonization.
- Any deviation from a perfect circular



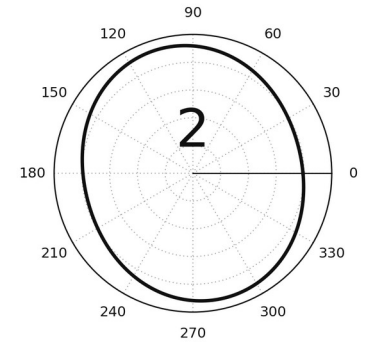


What is OOR (Out-of-Round)?

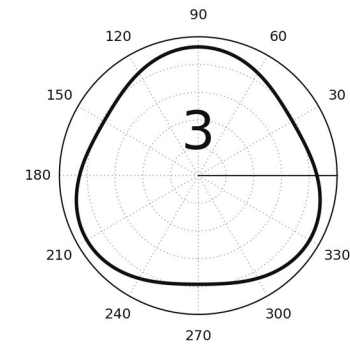
- Wheel out-of-roundness (OOR) is fundamentally a wear-driven phenomenon that **develops** through dynamic interaction between the wheel, vehicle, and track system.
- Initiated by a small defect then amplified by a resonance or repetitive excitation

HATCH

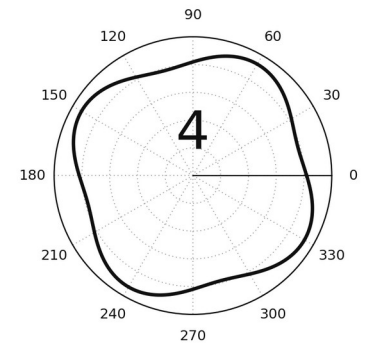
1st order — eccentric (still round)



2nd order — oval



3rd order — triangular



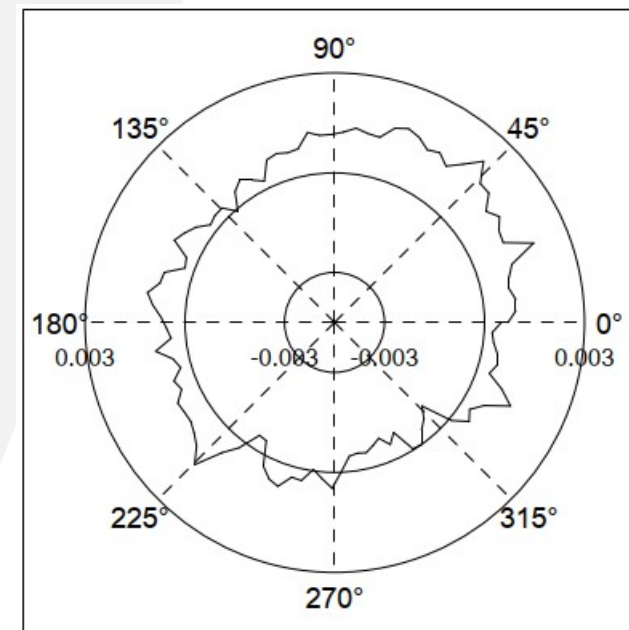
4th order — square



Case Study – Phase 1

Zarembski Technical Services LLC

- OOR condition exists in legacy fleets but is significantly more prevalent in the New fleet
- OOR wheels identified in:
 - 61% of New fleet-Series cars – 10% of repeat rate
 - 5% of Legacy Series cars – 0.3% of repeat rate
- Vehicle Design Differences:
 - Higher Carbody weight
 - Different suspension characteristics
 - Axle-to-axle distance: 88 inches





Case Study – Phase 2

Zarembski Technical Services LLC

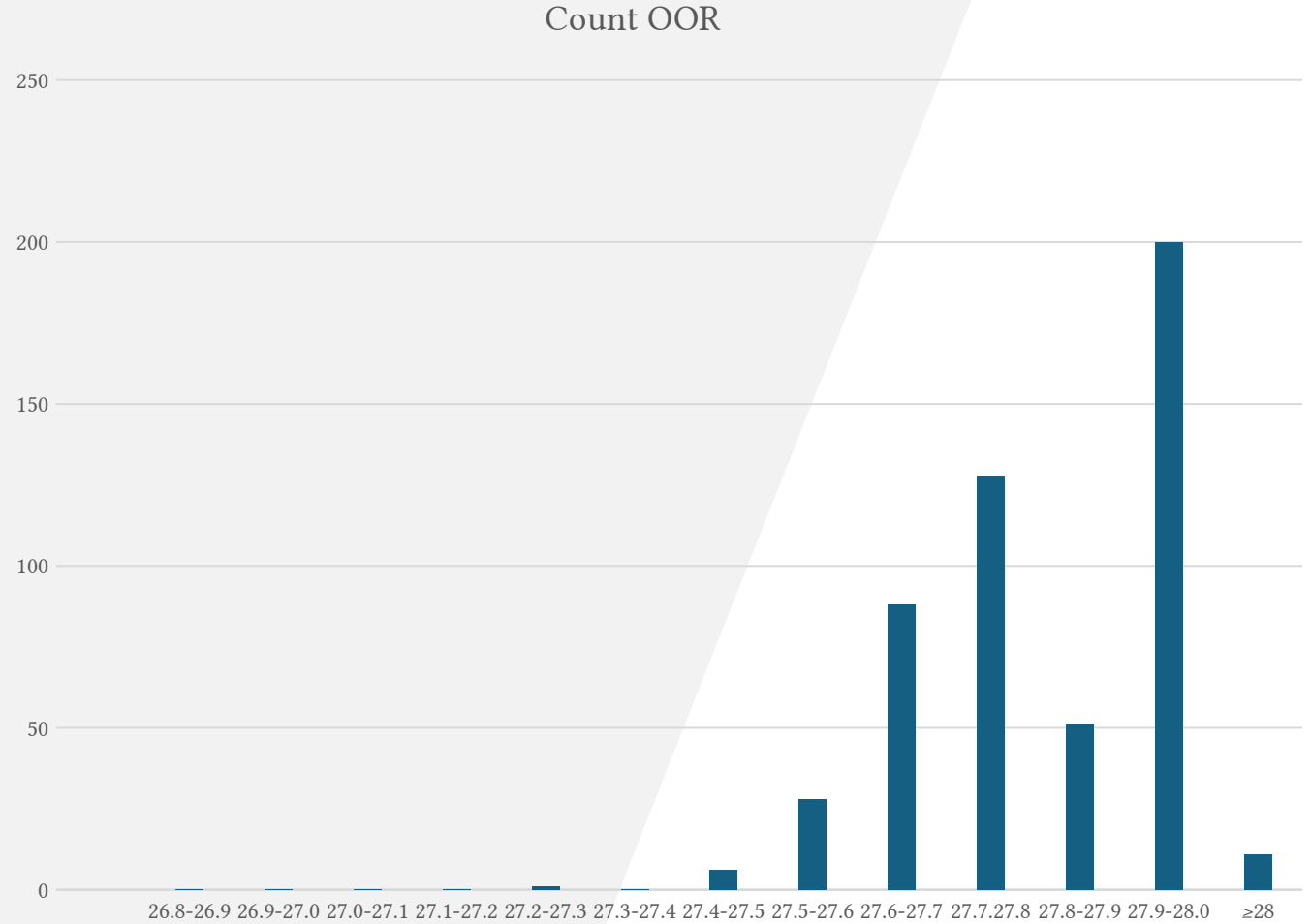
- Long-term wheel monitoring confirmed faster OOR growth on larger-diameter wheels (>27.5 in)
- Fourier analysis identified Order 3 OOR as the dominant wheel defect.
- Dynamic testing detected a 35-38 Hz longitudinal mode
- Results suggest a strong relationship between Order 3 OOR and a 35-38 Hz wheelset assembly mode



OOOR occurrence and Diameter relationship

HATCH

Oct 2018 – Feb 2020
(n=512)





Case Study – Phase 3



Zarembski Technical Services LLC

- Rail deflection measurements performed using a test train of new fleet
- The full truck model reproduced the previously observed:
 - Longitudinal and vertical modes at 35Hz
- Larger wheel diameters (28”) appear more susceptible to OOR growth than smaller worn wheels (<27.5”)
- Potential resonance mechanism linked to coincidence: Wheel circumference vs. Axle spacing



Simulation Model – Methodology

- Multibody system (MBS) formulation – the vehicle is abstracted as rigid and flexible bodies linked by joints and massless force elements
 - Joints impose constraint forces; springs, dampers and bushings impose applied forces
- Equations of motion are assembled automatically as a differential–algebraic system; wheel–rail contact enters as a kinematic constraint
- Three coupled subsystems – vehicle, wheel–rail contact, and track
- SIMPACK Rail module: quasi-elastic contact geometry, Hertzian normal force, FASTSIM tangential force

HATCH

1 Parameter collation

ZTS Phase III · WIA FEA · KHI drawings

2 MBS model build

bodies · joints · force elements

3 Wheel-rail interface

quasi-elastic geometry · Hertz · FASTSIM

4 Verification

equilibrium · eigenvalue · time-step

5 Validation

vs. Phase II measured shape + axlebox FFT

6 Parametric study

primary stiffness · loss factor

Schupp, G., Weidemann, C. and Mauer, L. (2004) Modelling the contact between wheel and rail within multibody system simulation. *Vehicle System Dynamics*, **41**(5), 349–364.



Measurement of OOR Wheels



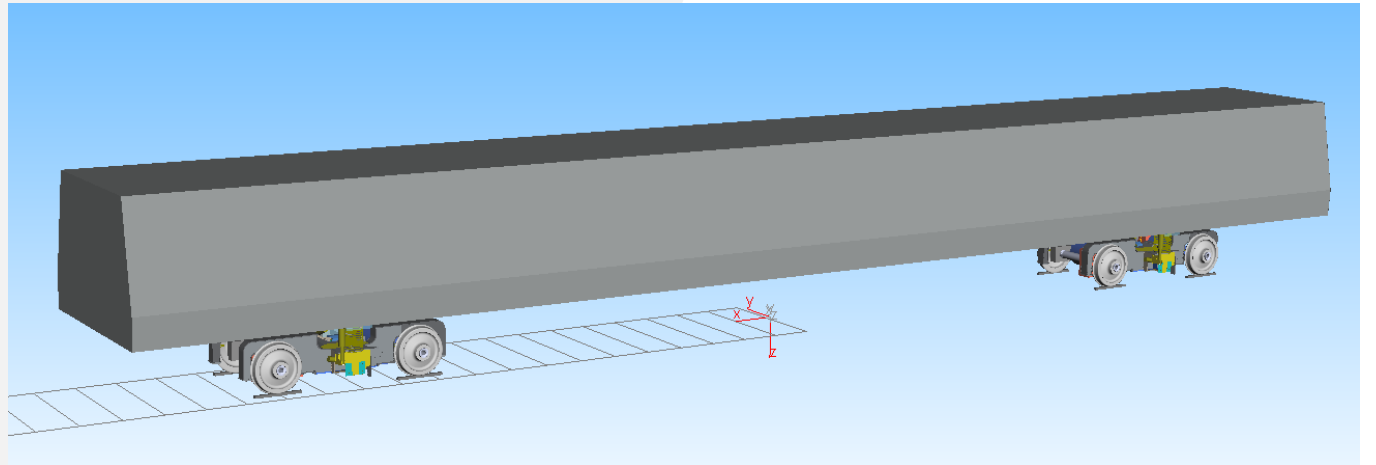
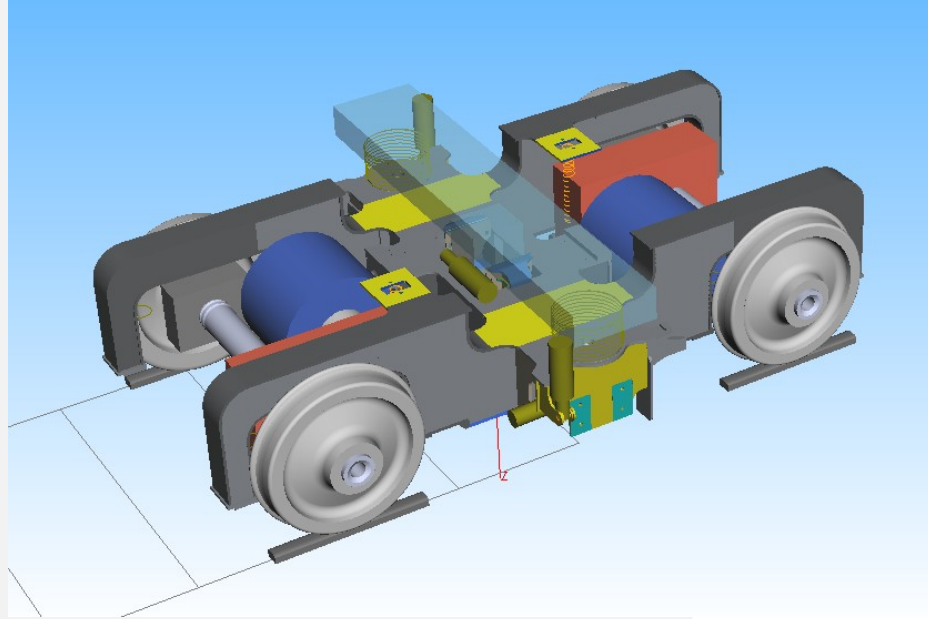
- Two measurement domains – wheel shape (geometry) and dynamic response (vibration)
- Shape: a displacement sensor records radial deviation while a rotation sensor records circumferential phase, giving the profile $f(x)$
 - The client measures on the Greenbelt wheel lathe – one point every 3.6° , 100 per revolution, orders resolved to 50
 - Phase I – 46 OOR wheels; Phase II – 16 wheels, monthly, 62,972 $\rightarrow$ 117,066 miles
- Vibration: tri-axial accelerometers at the bearing housing, ± 50 g, sampled at 1000 Hz
- Acceptance: 0.005 in new or newly trued, 0.015 in in service



Vehicle Dynamic Simulation Model

HATCH

- SIMPACK Model
 - Primary Suspension – Chevron Rubber
 - 2ry Suspension – Air spring + Lateral/Vertical Damper
 - Motor, Gear box, Journal Bearing, etc.
- 28 in wheelset, hollow axle, 88 in bogie wheel-base
- Wheel profile WHEEL_02-D6185-000 002-D
- Rail profile AREMA 115RE
- Tangent track
- Order-3 out-of-round, 0.1 mm amplitude
- Non-linear wheel/rail contact





Tangential Contact – the FASTSIM Algorithm



- Kalker's simplified theory: the elastic half-space is replaced by a bed of independent springs, so surface displacement is proportional to local traction
- Flexibility parameters L_1, L_2, L_3 are derived from the Kalker coefficients C_{11}, C_{22}, C_{23}
- The Hertzian ellipse is divided into strips along the rolling direction and elements across it (SIMPACK default 10×10)
- Traction is built up recursively from the leading edge, where it is zero
- Each element is in adhesion while the traction stays inside the Coulomb bound, and slips once it reaches it
- Integrating the traction over the patch gives F_x, F_y and M_z – roughly one thousandth the cost of Kalker's exact CONTACT

$$\mathbf{u}_\tau = \mathbf{L} \mathbf{p}_\tau, \quad \mathbf{L} = \text{diag}(L_1, L_2, L_3)$$

$$L_1 = \frac{8a}{3 C_{11} G}, \quad L_2 = \frac{8a}{3 C_{22} G}, \quad L_3 = \frac{\pi a^2}{4 c C_{23} G}, \quad c = \sqrt{ab}$$

$$\mathbf{w} = (\nu_x - \varphi y, \nu_y + \varphi x)$$

$$\mathbf{p}_\tau(x - \Delta x, y) = \mathbf{p}_\tau(x, y) + \Delta x \mathbf{L}^{-1} \mathbf{w}$$

$$|\mathbf{p}_\tau| \leq \mu p_z \text{ (adhesion); } \mathbf{p}_\tau = \mu p_z \frac{\mathbf{p}_\tau}{|\mathbf{p}_\tau|} \text{ (slip)}$$

Kalker, J.J. (1982) A fast algorithm for the simplified theory of rolling contact. *Vehicle System Dynamics*, **11**(1), 1–13.

Kalker, J.J. (1990) *Three-Dimensional Elastic Bodies in Rolling Contact*. Solid Mechanics and Its Applications, Vol. 2. Kluwer Academic Publishers, Dordrecht. ISBN 978-0-7923-0712-9.



Wheel–Rail Contact in the Coupled Model

HATCH

- SIMPACK resolves the contact in three steps
 - Locate the contact patch – position, size and shape
 - Compute the normal force
 - Compute the tangential force and moment
- Normal contact follows Hertz theory: the patch is an ellipse of semi-axes a and b under normal load P , with a semi-ellipsoidal pressure distribution
- Tangential forces follow Kalker's simplified non-linear rolling contact theory through FASTSIM
- Track: one lumped track/ballast element per wheelset, co-running with the wheelset along the track, carrying both rails
- This chain – OOR geometry → creepage → creep force → wear – is the mechanism under test

$$a = m \left[\frac{3\pi P (k_1 + k_2)}{4(A+B)} \right]^{1/3}, \quad b = n \left[\frac{3\pi P (k_1 + k_2)}{4(A+B)} \right]^{1/3} \quad (5)$$

$$p_z(x, y) = \frac{3P}{2\pi ab} \sqrt{1 - \left(\frac{x}{a}\right)^2 - \left(\frac{y}{b}\right)^2} \quad (6)$$

$$T_x = \iint_F X dx dy = -ab G C_{11} \nu_x \quad (7a)$$

$$T_y = \iint_F Y dx dy = -ab G (C_{22} \nu_y + \sqrt{ab} C_{23} \varphi) \quad (7b)$$

$$M_z = \iint_F (xY - yX) dx dy = -(ab)^{3/2} G (C_{23} \nu_y + \sqrt{ab} C_{33} \varphi) \quad (7c)$$

Hertz, H. (1882) Über die Berührung fester elastischer Körper. *Journal für die reine und angewandte Mathematik*, **92**, 156–171.

Liu, L.L., Liu, J.H. and Zuo, J.H. (2024) Multi-body dynamics modeling of heavy goods vehicle-rail interaction. *Open Journal of Applied Sciences*, **14**(7), 1715–1722.



OOR Wheel Model



- The non-circularity is described by a harmonic radius perturbation superimposed on the nominal rolling radius
- ΔR is the radius deviation, A its amplitude, n the polygon order, θ the wheel rotation angle, θ_0 the initial phase and ω the angular velocity
- Changing order, amplitude, initial phase and angular velocity simulates any OOR condition; in SIMPACK the deviation is indexed on wheel angular position, so it repeats once per revolution
- Baseline: $n = 3$, $A = 0.1$ mm zero-to-peak, giving a longitudinal creepage perturbation $\Delta r/r = 0.028$ %

$$\Delta R = A \sin(n\theta + \theta_0) \quad (8a)$$

$$R(\theta) = R - \Delta R \quad (8b)$$

$$\theta(t) = \theta(t-1) + \omega \Delta t \quad (8c)$$



Verification – the rigid-track baseline

Before any elastic-track result is quoted, the same excitation is run on a rigid track. Converting the vertical wheelset acceleration to displacement by dividing by $\omega^2 = (2\pi f_0)^2$:

Speed	f_0 (Hz)	$\hat{a}_z$ (m/s ²)	$\hat{a}_z/\omega^2$ (mm)	vs imposed 0.1 mm
45 mph	26.9857	2.857	0.0994	0.99
59 mph	35.3820	5.102	0.1032	1.03
65 mph	38.9802	6.274	0.1046	1.05
70 mph	41.9786	7.311	0.1051	1.05

The rigid wheelset traces the imposed out-of-roundness to within 5% at every speed. Two things follow: the 0.1 mm reaches the contact, and the model contributes no dynamics of its own. Every magnification quoted later is measured against this baseline.



Verification – the forcing is kinematic

The effective rolling radius, recovered as $r_{\text{eff}} = nV/2\pi f_0$ from the measured peak:

Speed	45	50	55	60	65	70 mph
Rigid track	0.35593	—	—	0.35592	0.35592	0.35592 m
Elastic track	0.35692	0.35692	0.35692	0.35693	0.35693	0.35693 m

Constant to five decimals within each model, and a $2\times$ change in track stiffness moves it by 0.28%. The excitation frequency is therefore purely kinematic, $f_0 = nV/2\pi r$; track properties set the *gain*, not the frequency.



Formation Mechanisms of OOR

HATCH

- Three competing mechanisms
 - Fixed wavelength – periodic impact at a constant spatial interval
 - Fixed frequency – a structural resonance of the wheel–rail system
 - Non-uniform initial hardness from manufacture or work hardening
- Rail pinned–pinned frequency sets a candidate fixed frequency
- Metal-cutting chatter analogy: the number of marks per revolution is the ratio of self-excitation frequency to wheel rotation frequency
 - A non-integer ratio closes after m revolutions, giving $N = mN_0 - 1$ corners
- **New fleet at 58 mph: $f_\omega = 11.6$ Hz, so order 3 falls at 34.8 Hz – the DF11 band**

$$f_{pp} = \frac{\pi}{2l^2} \sqrt{\frac{EI}{m}} \quad (3)$$

$$N = \frac{60f}{n} = \frac{f_z}{f_w} \quad (4)$$

$$N_0 + \varepsilon = \frac{60f}{n} = \frac{f_z}{f_w}, \quad -1 < \varepsilon < 1 \quad (5)$$

$$N = m(N_0 + \varepsilon) = mN_0 - 1 \quad (6)$$

$$f_\omega = \frac{v}{2\pi R} \quad (7)$$

Nielsen, J.C.O. and Johansson, A. (2000) Out-of-round railway wheels — a literature survey. *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit*, **214**(2), 79–91.



Ruling Out Rail Pinned–Pinned Resonance

HATCH

- Equation (3) of Shan & Song gives the first mode of one rail bay, simply supported between fasteners
- Lateral bending needs the WEAK-axis inertia – 115RE $I_y = 10.8 \text{ in}^4$, not $I_x = 65.6 \text{ in}^4$
- 30 in pitch, 115RE at 57.0 kg/m: 348 Hz lateral, 858 Hz vertical
- Method validated – CHN60 at 0.60–0.63 m gives 533–588 Hz, overlapping Shan & Song's reported 559–602 Hz
- Pin-pin demands orders 25–39. The fleet has an order-3 problem – wrong band by a factor of x10

Speed	Order at 348 Hz	Order at 858 Hz
45 mph	39	95
58 mph	30	74
70 mph	25	61



Pin–Pin Is Not the Fastener Passing Frequency

HATCH

	Formula	At 60 mph	Behaviour
Pin-pin resonance	$(\pi / 2l^2) \sqrt{(EI/m)}$	348 Hz	Fixed – does not move with speed
Fastener passing	V / l	35.2 Hz	Kinematic – scales with speed

- Both derive from the same support pitch l
- Pin-pin is a structural resonance at 348 Hz, far outside the excitation band
- Fastener passing is a kinematic forcing frequency, squarely inside it
- **Eliminating pin-pin does not eliminate the fastener pitch – it redirects attention to the passing frequency**

Nielsen, J.C.O. and Johansson, A. (2000) Out-of-round railway wheels — a literature survey. *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit*, **214**(2), 79–91.



The Order-3 Wavelength IS the Fastener Pitch

HATCH

- Fastener passing and order-3 OOR track each other at a constant 2.3 % offset at every speed
- Both are V divided by a length – and the two lengths are nearly the same
 - $\lambda_3 = \pi D / 3 = 0.7448 \text{ m}$ versus DF pitch = 0.7620 m
- There is no special speed. This is a fixed-WAVELENGTH mechanism, not a resonance

Speed	Fastener passing	Order-3 OOR	Offset
45 mph	26.4 Hz	27.0 Hz	2.3 %
55 mph	32.3 Hz	33.0 Hz	2.3 %
58 mph	34.0 Hz	34.8 Hz	2.3 %
60 mph	35.2 Hz	36.0 Hz	2.3 %
70 mph	41.1 Hz	42.0 Hz	2.3 %



The 27.5-Inch Cut-Off, From Geometry Alone

HATCH

Wheel dia.	$\lambda_3 = \pi D / 3$	vs 0.762 m pitch	C / pitch	n - 3	m = 1 / ε
28.2 in	0.7501 m	-1.6 %	2.953	0.047	21.3 rev
28.0 in	0.7448 m	-2.3 %	2.932	0.068	14.7 rev
27.6 in	0.7341 m	-3.7 %	2.890	0.110	9.1 rev
27.5 in	0.7315 m	-4.0 %	2.880	0.120	8.3 rev
27.0 in	0.7182 m	-5.8 %	2.827	0.173	5.8 rev
26.6 in	0.7075 m	-7.2 %	2.786	0.214	4.7 rev

- As the wheel wears λ_3 shrinks, while the DF fastener pitch stays fixed at 0.762 m
- The last column is the Integral Multiple Condition — Iwnicki, Nielsen & Tao (2023), Vehicle System Dynamics 61(7):1785–1828, §8.1: a polygon of order n closes around the wheel only when the ratio of circumference to wavelength, C / λ , is a whole number
- Iwnicki applies it with $\lambda = V / f$ from a fixed-frequency mechanism, and observes that the condition is met several times within one wheel life — so susceptibility is banded in diameter, not monotonic. Here λ is instead the fixed 0.762 m fastener pitch, so the band follows from geometry with no frequency involved
- Within about 4 % of the pitch from 27.6 to 28.2 in. The -4% line falls at exactly 27.5 in, below that the imprint stops reinforcing
- **In line with 27.5-inch observation — from geometry alone. No resonance required.**



Why Only the New fleet? — Still Open



Fleet	Wheelbase	WB / C at 28 in	Departure from 1.0
Legacy 1xxx series	87.00 in	0.98903	1.10 %
Legacy 2xxx/3xxx/4xxx series	90.55 in	1.02939	2.94 %
Legacy 5xxx/6xxx series	87.00 in	0.98903	1.10 %
New fleet	88.00 in	1.00040	0.04 %

- Every fleet runs 28 in wheels on the same 30 in DF fastener pitch. Only the wheelbase differs — Phase III, Table 10. Wheel circumference at 28 in is 87.965 in.
- **VERIFIED: the New fleet is the only fleet whose bogie wheelbase equals its own wheel circumference. 88.00 in against 87.97 in, a departure of 0.04 %. No other fleet is within 1.1 %.**
- **HYPOTHESIS, not established — wheelbase filtering:**
 - Two contact points a wheelbase apart feed the same bogie. Order-3 OOR imprints a wavelength of $C/3$, so the two axles act in phase when $3 \cdot WB/C$ is a whole number: 3.0012 for the New fleet, against 2.967 and 3.088 for the legacy fleets.
 - It is not proven. It requires the order-3 phase on the two wheels to be related to each other, which this study has not established. And for a track-fixed input at the 0.762 m pitch the same filter favours the legacy fleet, which is unaffected.



What the Fleet Data Rules Out

- The Legacy fleet uses the same chevron primary as the New fleet
- Its longitudinal stiffness is 139,xxx lb/in – within 1.2 % of the New fleet's 138,xxx lb/in
- That fleet does not have the OOR problem
- Still unique to the New fleet: separated motor-gear, softest primary vertical of any fleet at 7,5xx lb/in, 63° flange angle (shifting to 70°), heaviest car at 84,xxx lb
- **Primary longitudinal stiffness cannot be the discriminator – the fleet that shares it is unaffected**

HATCH

Primary longitudinal	lb/in
1xxx (rubber bushing)	106,xxx
5xxx/6xxx (rubber bushing)	115,xxx
2xxx/3xxx/4xxx (chevron)	139,xxx
New fleet (chevron)	138,xxx



Where This Leaves the Investigation

HATCH

- ESTABLISHED
 - Pin-pin resonance ruled out — wrong band by a factor of ten
 - Order-3 wavelength equals the DF fastener pitch to 2.3 %, at every speed
 - The 27.5-inch diameter cut-off follows from wheel geometry against a fixed fastener pitch
- STILL OPEN
 - Why the New fleet — the wheelbase/circumference coincidence is verified to 0.04 %, but no mechanism for it has been established
 - The softest primary vertical of any fleet couples to P2 — WIA place it at 58 Hz on DF track and 35 Hz on ballast
- RULED OUT
 - Rail pinned-pinned resonance; primary longitudinal stiffness as the fleet discriminator;
 - Fixed-wavelength, not fixed-frequency: geometry sets the order and the diameter band. What selects the fleet is still open.



Discovery 1 — Geometry Sets the Diameter Band



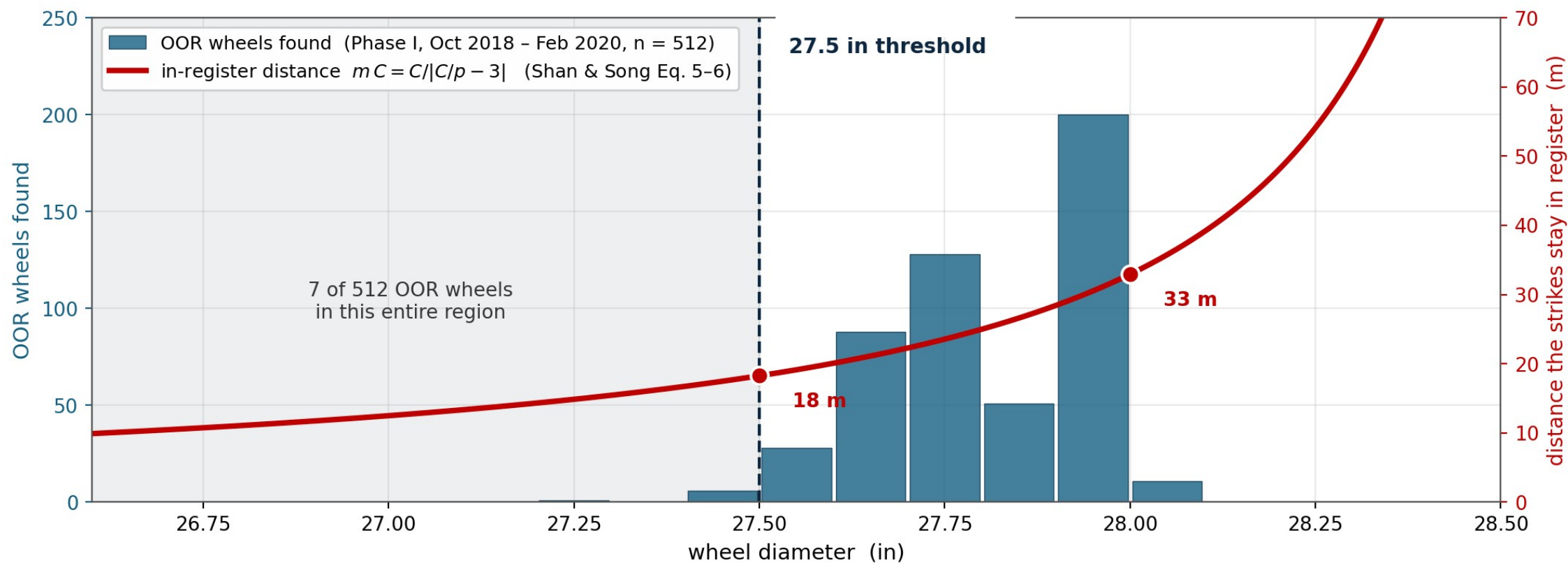
Wheel dia.	C (m)	$N = C / p$	$\varepsilon = N - 3$	$m = 1 / \varepsilon $	$m \cdot C$
28.2 in	2.2503	2.9531	-0.047	21.3 rev	48 m
28.0 in	2.2343	2.9322	-0.068	14.7 rev	33 m
27.6 in	2.2024	2.8903	-0.110	9.1 rev	20 m
27.5 in	2.1944	2.8798	-0.120	8.3 rev	18 m
27.0 in	2.1545	2.8274	-0.173	5.8 rev	12 m

- Fastener passing divided by wheel rotation is $(V/p) / (V/C) = C / p$ — the speed cancels, so the ratio is fixed by geometry alone. At 28 in it is 2.9322, not the whole number 3
- Shan & Song (2018), Eq. 5–6: write the ratio as $N_0 + \varepsilon$ about the nearest whole number. The strike pattern then slips one full lobe spacing after $m = 1 / |\varepsilon|$ wheel revolutions
- At 28.0 in the fastener strikes stay in register with the three lobes for 14.7 revolutions — 33 m of track. At 27.5 in only 8.3 revolutions, 18 m. A factor of 1.8 across the client's observed threshold, and 3.7 across the wear range
- Only the wheel diameter changes. The 0.762 m fastener pitch is fixed, and is the same under every fleet
- The diameter band follows from two lengths — πD and the fastener pitch. No resonance, no vehicle parameter, nothing fitted.



Discovery 1 – The Band Against 512 Measured Wheels

HATCH

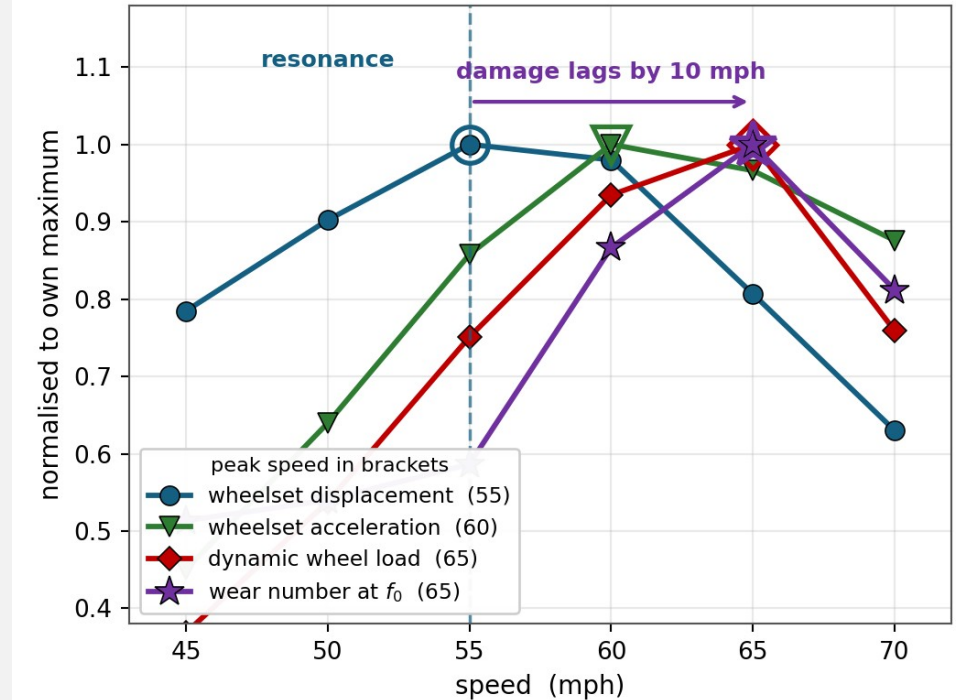


- Counts binned from the Phase I dataset, Oct 2018 – Feb 2020: only 7 of 512 OOR wheels lie below 27.5 in — 1.4 %. It is the sharpness of that cut-off, not the trend above it, that matches a geometric threshold. These are counts, not rates.

Discovery 2 — Damage Peaks Above Resonance

HATCH

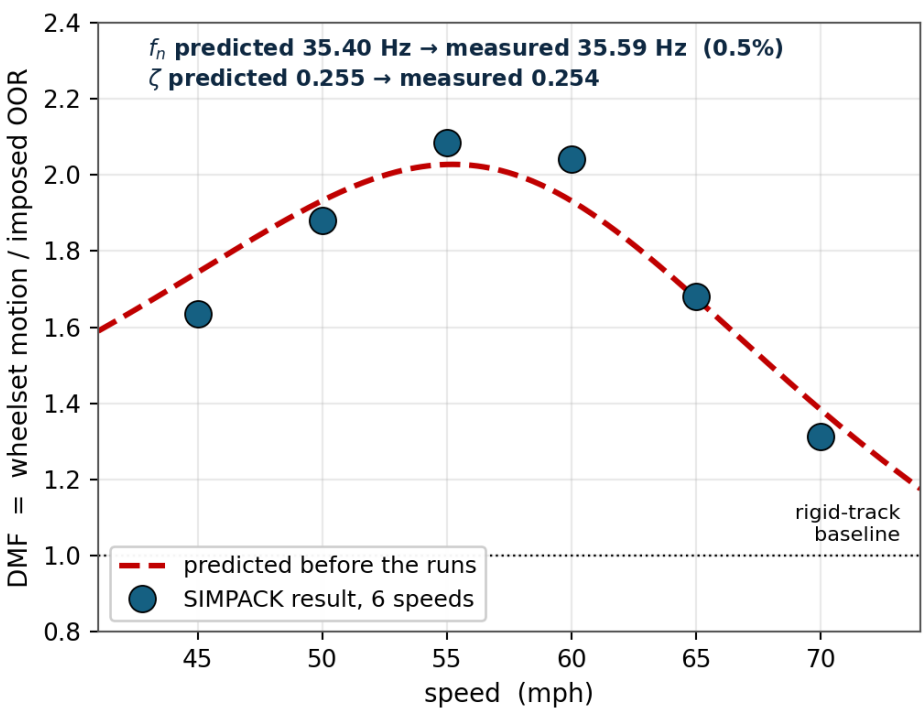
- Tunnel sweep, $c_{z0} = 108.6 \text{ MN/m}$, six speeds 45–70 mph: vertical resonance 35.6 Hz, order 3 coincides at 59.5 mph
- All four response measures peak — and they peak at different speeds:
 - wheelset displacement at 55 mph — that is the resonance itself
 - wheelset acceleration at 60 mph
 - dynamic wheel load at 65 mph, $\pm 26.1 \%$ of static
 - OOR-synchronous wear number at 65 mph, $+95 \%$ on its 45 mph value
- Wear number is $|T \cdot \gamma|$ — creepage scales with velocity, displacement does not, so the damage measures lag the displacement resonance by a full 10 mph
- The speed at which the track resonates is not the speed at which the polygon grows fastest. A speed restriction set at the resonance would target the wrong point.



Discovery 3 — Predicted First, Then Confirmed



- Two completed sweeps — $c_{z0} = 150$ and 75 MN/m — returned $f_n = 43.34$ Hz and 29.42 Hz
- Those two points alone fix the placement law $f_n = 29.42 \sqrt{(c_{z0} / 75 \text{ MN/m})}$ with 8% mass scatter explaining the 4% miss at 150MN/m
- $c_{z0} = 108.6$ MN/m was then chosen to place f_n at 35.4 Hz, and the whole six-speed DMF curve was predicted before any of the runs were executed
- Measured over $45\text{--}70$ mph: $f_n = 35.59$ Hz — 0.5 % from prediction — with $\zeta = 0.254$ against 0.255 predicted
- Order 3 therefore coincides with the vertical mode at 59.5 mph



$$DMF(f) = \frac{1}{\sqrt{(1 - (f/f_n)^2)^2 + (2\zeta f/f_n)^2}}$$

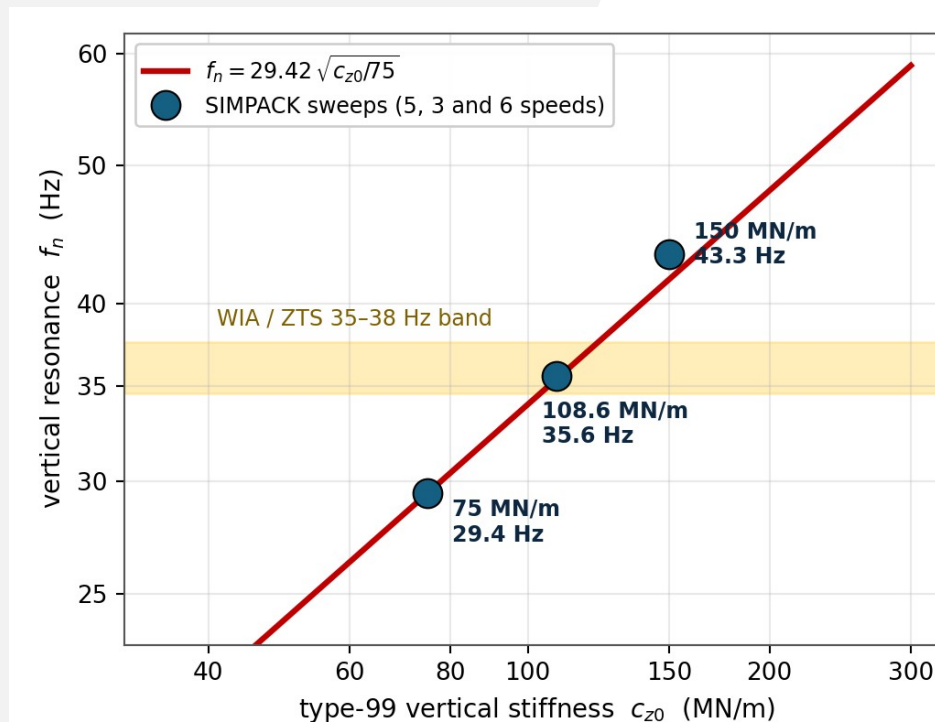
c_{z0}	Speeds fitted	f_n	ζ	M_{eff}
150 MN/m	45, 55, 60, 65, 70	43.34 Hz	0.251	2,023 kg
75 MN/m	45, 55, 65	29.42 Hz	0.255	2,196 kg



Discovery 3 — An Auditable Verification Chain

HATCH

- Rigid-track control returns $DMF = 1.00 \pm 0.03$ at every speed — the imposed 0.1 mm reaches the contact and the model adds no dynamics of its own
- r_{eff} constant to five decimals across a $2\times$ stiffness change — the forcing is purely kinematic, $f_0 = nV / 2\pi r$
- Effective vertical modal mass stable at 2,023 / 2,196 / 2,172 kg over three stiffnesses — a model property, not a curve fit
- Independent pipeline check: SIMPACK-side FFT 32.8895 Hz / 9.1725 m/s^2 against 32.8909 Hz / 9.175 m/s^2 by complex demodulation



Why a Fixed-Speed Order Swap Cannot Test the Mechanism



OOR order n	f_0 at 59 mph	f_0 / f_n	DMF — predicted, no run needed	Speed that puts n on resonance
2	23.5 Hz	0.66	1.53	89.3 mph
3	35.3 Hz	0.99	1.98	59.5 mph
4	47.0 Hz	1.32	1.00	44.6 mph
5	58.8 Hz	1.65	0.52	35.7 mph
6	70.6 Hz	1.98	0.32	29.8 mph
8	94.1 Hz	2.64	0.16	22.3 mph

- At one speed every order sits at a different forcing frequency, $f_0 = nV / 2\pi r$, so the six runs would sample six points on the same resonance curve rather than test six orders
- The whole DMF column above is written from the calibrated SDOF fit — $f_n = 35.59$ Hz, $\zeta = 0.254$ — with no simulation at all, and the peak DMF the damping allows is 2.04
- Order 3 leads by $2\times$ because $cz_0 = 108.6$ MN/m was chosen to place it on resonance at 59.5 mph. The result is built into the setup.
- The non-circular test holds f_0 fixed and varies speed instead — last column — so the track dynamics are identical, and any difference is the order's own contribution





Three Findings

HATCH

- **1. Order-3 OOR is a fixed-wavelength mechanism, not a resonance**
 - The order-3 wavelength $\lambda_3 = \pi D/3 = 0.7448$ m and the 0.762 m DF fastener pitch track each other at a constant 2.3 % offset at every speed — there is no special speed. As the wheel wears the two diverge, past 5 % below 27.5 in, which is the observed cut-off.
- **2. The 35-38Hz track resonance acts as an amplifier, not the root cause**
 - A 35-38 Hz vertical track resonance exists in the SIMPACK model and can amplify OOR excitation, but it appears to be an amplifier whose frequency is controlled by track stiffness rather than the root cause of OOR.
- **3. Damage peaks 10 mph above the resonance**
 - The worst wheel damage condition occurs around 65 mph, above the resonance speed, because wheel load and wear continue increasing after displacement resonance has passed.

Mitigation 1: Wheel diameter management



Re-profiling strategy	Life spent in the 27.5–28.0 in window	OOR exposure	Wheel-life cost
Small successive cuts (current practice)	most of the wheel's service life	high	baseline
First cut taken from 28.0 straight below 27.5 in	none	low	17% on the first turning

- Every wheel of New fleet enters service at 28 in., and small successive cuts keep it in the 27.5–28.0 in. band for extended time
- Examination records: 505 of 512 OOR wheels — 98.6 % — were found at 27.5 in. or above (Phase I, Oct 2018 – Feb 2020)
- The mechanism is the fastener-pitch imprint: at 28.0 in. the order-3 pattern holds for 14.7 rev before it drifts; at 27.0 in. only 5.8 rev
- Making the first re-profiling cut deep enough to clear 27.5 in. in one pass skips the band entirely, at a cost of about 0.5 in of wheel life on that first turning
- Trialable on a subset of the fleet: cheap, reversible, and measurable against the same lathe records